

Lemma: $n \in \mathbb{Z}_{\geq 1}$ 且 $\sigma \in \mathfrak{S}_n$. 则有: $\sigma^{n!} = \text{id}_{\{1, \dots, n\}}$

Proof: $\because n \in \mathbb{Z}_{\geq 1} \quad \therefore |\mathfrak{S}_n| = n! \in \mathbb{Z}_{\geq 1} \quad \because \mathfrak{S}_n$ 是有限群且 $\sigma \in \mathfrak{S}_n$

$\therefore \text{ord}(\sigma) \in \mathbb{Z}_{\geq 1}$ 且 $\text{ord}(\sigma) \mid |\mathfrak{S}_n| \quad \therefore \text{ord}(\sigma) \mid n!$

$\because \mathfrak{S}_n$ 是群, $\sigma \in \mathfrak{S}_n$, $\text{ord}(\sigma) \in \mathbb{Z}_{\geq 1}$, $n! \in \mathbb{Z}_{\geq 1}$, $\text{ord}(\sigma) \mid n!$

$\therefore \sigma^{n!} = \text{id}_{\mathfrak{S}_n} = \text{id}_{\{1, \dots, n\}} \quad \square$

Lemma: ~~非~~ X 是非空集合, 对 $\forall \sigma \in \mathfrak{S}_X$, 有:

$$\text{supp}(\sigma^{-1}) = \text{supp}(\sigma)$$

Proof: $\because \sigma \in \mathfrak{S}_X \quad \therefore \text{supp}(\sigma) = \{x \in X \mid \sigma(x) \neq x\} \subseteq X$

$\because (\mathfrak{S}_X, \circ)$ 是群, $\sigma \in \mathfrak{S}_X \quad \therefore \sigma^{-1} \in \mathfrak{S}_X$

$$\therefore \text{supp}(\sigma^{-1}) = \{x \in X \mid \sigma^{-1}(x) \neq x\} \subseteq X$$

对 $\forall x \in \text{supp}(\sigma)$, 有: $x \in X$ 且 $\sigma(x) \neq x$. 假设 $x \notin \text{supp}(\sigma^{-1})$.

则 $\sigma^{-1}(x) = x \quad \therefore \sigma(x) = \sigma(\sigma^{-1}(x)) = x$ 矛盾. $\therefore x \in \text{supp}(\sigma^{-1})$

$$\therefore \text{supp}(\sigma) \subseteq \text{supp}(\sigma^{-1})$$

对 $\forall x \in \text{supp}(\sigma^{-1})$, 有: $x \in X$ 且 $\sigma^{-1}(x) \neq x$. 假设 $x \notin \text{supp}(\sigma)$

则 $\sigma(x) = x. \quad \therefore \sigma^{-1}(x) = \sigma^{-1}(\sigma(x)) = x$ 矛盾. $\therefore x \in \text{supp}(\sigma)$

$$\therefore \text{supp}(\sigma^{-1}) \subseteq \text{supp}(\sigma)$$

$$\therefore \text{supp}(\sigma^{-1}) = \text{supp}(\sigma) \quad \square$$

定义 (m -循环或轮换) X 是非空集合, $a_1, \dots, a_m$ 是 X 中相异的元素 ($m \in \mathbb{Z}_{\geq 1}$). 方便起见, 取同余类将 a_i 的下标视同 $\mathbb{Z}/m\mathbb{Z}$ 的元素. 具有以下形式的置换:

$$\begin{aligned} \sigma: X &\longrightarrow X \\ \sigma(a_i) &= a_{i+1}, \quad i \in \mathbb{Z}/m\mathbb{Z} \\ \sigma(x) &= x, \quad x \in X \setminus \{a_1, a_2, \dots, a_m\} \end{aligned}$$

称为 X 上的 m -循环或轮换, 也记为 $\sigma = (a_1 \dots a_m) \in \mathcal{S}_X$.

对于 $m=2$ 的情形, 相应的 $(a_1 a_2)$ 称为元素 a_1 和 a_2 的对换.

Lemma: X 是非空集合, $a_1, \dots, a_m \in X$ 是 X 中相异的元素 ($m \in \mathbb{Z}_{\geq 1}$)

定义映射 $\sigma: X \longrightarrow X$. 则有: $\sigma \in \mathcal{S}_X$.

$$\begin{aligned} \sigma(a_i) &= a_{i+1}, \quad i \in \mathbb{Z}/m\mathbb{Z} \\ \sigma(x) &= x, \quad x \in X \setminus \{a_1, \dots, a_m\} \end{aligned}$$

Proof: 显然 $\sigma(X) \subseteq X$.

若 $a_i = a_j$, 则 $i \equiv j \pmod{m}$ $\therefore i+1 \equiv j+1 \pmod{m}$

$\therefore a_{i+1} = a_{j+1}$ $\therefore \sigma(a_i) = a_{i+1} = a_{j+1} = \sigma(a_j)$ $\therefore \sigma$ 是 $X \rightarrow X$ 的映射.

对 $\forall x, y \in X$. 若 $\sigma(x) = \sigma(y)$, 则有:

(i) 如果 $x, y \in \{a_1, \dots, a_m\}$. 则 $x = a_i$, $y = a_j$. $\therefore \sigma(x) = a_{i+1}$.

$\sigma(y) = a_{j+1}$ $\therefore a_{i+1} = a_{j+1}$ $\therefore i+1 \equiv j+1 \pmod{m}$ $\therefore i \equiv j \pmod{m}$

$\therefore i \equiv j \pmod{m}$ $\therefore a_i = a_j$ $\therefore x = y$.

(ii) 如果 $x, y \in X \setminus \{a_1, \dots, a_m\}$. 则 $x = \sigma(x) = \sigma(y) = y$

(iii). 如果 $x \in \{a_1, \dots, a_m\}$ 且 $y \in X \setminus \{a_1, \dots, a_m\}$. 设 $x = a_i$

$$\therefore y = \sigma(y) = \sigma(x) = \sigma(a_i) = a_{i+1} \quad \therefore y = a_{i+1} \in \{a_1, \dots, a_m\}.$$

矛盾. $\therefore$ 这种情况不可能出现.

同理可证 $x \in X \setminus \{a_1, \dots, a_m\}$ 且 $y \in \{a_1, \dots, a_m\}$ 不可能出现.

$\therefore x = y$. $\therefore \sigma$ 是 $X \rightarrow X$ 的单射.

对 $\forall y \in X$. 若 $y \in X \setminus \{a_1, \dots, a_m\}$ 则 $\sigma(y) = y$

若 $y \in \{a_1, \dots, a_m\}$. 则设 $y = a_i$. $\therefore \sigma(a_{i-1}) = a_i = y$.

$\therefore \sigma$ 是 $X \rightarrow X$ 的满射. $\therefore \sigma$ 是 $X \rightarrow X$ 的双射. $\therefore \sigma \in \mathcal{G}_X$

□

Lemma: X 是非空集合, 则 X 上的 1-循环 $= \text{id}_X$.

Proof: $\mathbb{Z}/1\mathbb{Z} = \{1\}$. 若 $\sigma \in \mathcal{G}_X$ 是 1-循环, 则

$$\sigma(a_1) = a_1, \quad \sigma(x) = x, \quad x \in X \setminus \{a_1\}. \quad \therefore \sigma = \text{id}_X. \quad \square$$

Lemma: $(a_1 \dots a_m) = (a_2 \dots a_m a_1)$

Proof: 记 $\sigma = (a_1 \dots a_m) \in \mathcal{G}_X$, $\tau = (a_2 \dots a_m a_1) \in \mathcal{G}_X$.

$\therefore$ 对 $\forall x \in X \setminus \{a_1, \dots, a_m\}$, 有: $\sigma(x) = x$, $\tau(x) = x$.

$$\therefore \sigma(x) = \tau(x).$$

$$\therefore \sigma(a_1) = a_2, \quad \tau(a_1) = a_2.$$

$$\therefore \sigma(a_1) = \tau(a_1)$$

$$\sigma(a_2) = a_3, \quad \tau(a_2) = a_3.$$

$$\therefore \sigma(a_2) = \tau(a_2)$$

$$\sigma(a_m) = a_1, \quad \tau(a_m) = a_1$$

$$\therefore \sigma(a_m) = \tau(a_m)$$

$$\therefore \sigma = \tau$$



Lemma: $\xi = (a_1 \dots a_m) \in \mathcal{G}_X$. 则有: ~~对~~ $\forall k \in \mathbb{Z}_{\geq 1}$,

$$\xi^k: X \longrightarrow X$$

$$a_i \mapsto a_{i+k}.$$

对 $\forall x \in X \setminus \{a_1, \dots, a_m\}$, 有: $x \mapsto x$

Proof: 当 $k=1$ 时, $\xi^k = \xi' = \xi$ 结论成立.

假设当 $k=n \in \mathbb{Z}_{\geq 1}$ 时, 结论成立. 则 $k=n+1$ 时,

$$\xi^k = \xi^{n+1} = \xi^n \circ \xi' = \xi^n \circ \xi$$

$$\therefore \xi^k(a_i) = \xi^n(\xi(a_i)) = \xi^n(a_{i+1}) = a_{(i+1)+n} = a_{i+(n+1)}$$

$$= a_{i+(n+1)} = a_{i+k}$$

对 $\forall x \in X \setminus \{a_1, \dots, a_m\}$, 有: $\xi^k(x) = \xi^n(\xi(x)) = \xi^n(x) = x$

$\therefore$ 当 $k=n+1$ 时结论成立. $\therefore$ 对 $\forall k \in \mathbb{Z}_{\geq 1}$, 结论成立.



Lemma: $\xi = (a_1 \cdots a_m) \in \mathcal{S}_X$, 则有: $\text{ord}(\xi) = m$ (其中 $m \in \mathbb{Z}_{\geq 1}$)

Proof: $\because \xi^m: X \longrightarrow X$

$$a_i \longmapsto a_{i+m}$$

又 $\forall x \in X \setminus \{a_1, \dots, a_m\}$, 有: $x \mapsto x$

$$\therefore \xi^m(a_i) = a_{i+m} = a_i, \quad \text{又} \forall x \in X \setminus \{a_1, \dots, a_m\}, \text{有: } \xi^m(x) = x$$

$$\therefore \xi^m = \text{id}_X$$

又 $\forall k \in \mathbb{Z}_{\geq 1}$ 且 $1 \leq k < m$, 有: $\xi^k: X \longrightarrow X$

$$a_i \longmapsto a_{i+k}$$

又 $\forall x \in X \setminus \{a_1, \dots, a_m\}$, 有: $x \mapsto x$.

$$\therefore \xi^k(a_1) = a_{1+k} = a_{k+1} \quad \because 1 \leq k < m \quad \therefore m \nmid k$$

$$\therefore m \nmid (k+1) - 1 \quad \therefore k+1 \not\equiv 1 \pmod{m}. \quad \therefore \xi^k(a_1) = a_{k+1} \neq a_1$$

$$\therefore \xi^k \neq \text{id}_X.$$

$$\therefore \text{ord}(\xi) = m \quad \square$$

定义(两个循环不交) 给定集合 X 上的循环 $(a_1 \cdots a_n)$ 和 $(b_1 \cdots b_m)$, 若 $\{a_1, \dots, a_n\} \cap \{b_1, \dots, b_m\} = \emptyset$, 则称这两个循环不交.

Lemma: 设循环 $\sigma, \tau \in \mathbb{S}_X$ 不交, 则 $\sigma\tau = \tau\sigma$

Proof: 设 $\sigma = (a_1 \cdots a_n)$, $\tau = (b_1 \cdots b_m)$. 其中

$$\{a_1, \dots, a_n\} \cap \{b_1, \dots, b_m\} = \emptyset.$$

则有: $\because \sigma, \tau \in \mathbb{S}_X \quad \therefore \sigma\tau \in \mathbb{S}_X, \tau\sigma \in \mathbb{S}_X.$

~~$\forall x \in X \setminus \{a_1, \dots, a_n, b_1, \dots, b_m\}$~~

$\forall x \in X \setminus \{a_1, \dots, a_n, b_1, \dots, b_m\}$, 有: $\sigma\tau(x) = \sigma(\tau(x)) = \sigma(x) = x$. $\tau\sigma(x) = \tau(\sigma(x)) = \tau(x) = x$. $\therefore \sigma\tau(x) = \tau\sigma(x)$

$$\sigma\tau(a_i) = \sigma(\tau(a_i)) = \sigma(a_i) = a_{i+1}$$

$$\tau\sigma(a_i) = \tau(\sigma(a_i)) = \tau(a_{i+1}) = a_{i+1} \quad \therefore \sigma\tau(a_i) = \tau\sigma(a_i)$$

$$\sigma\tau(b_i) = \sigma(\tau(b_i)) = \sigma(b_{i+1}) = b_{i+1}$$

$$\tau\sigma(b_i) = \tau(\sigma(b_i)) = \tau(b_{i+1}) = b_{i+1} \quad \therefore \sigma\tau(b_i) = \tau\sigma(b_i)$$

$$\therefore \sigma\tau = \tau\sigma. \quad \square$$

Lemma: X 是集合, $f: X \rightarrow X$ 是一个双射, $x \in X$. $\exists r \in \mathbb{Z}_{\geq 1}$

s.t. $f^r(x) = x$. 令 $\ell = \min\{n \in \mathbb{Z}_{\geq 1} \mid f^n(x) = x\}$. 则有:

① $x, f(x), f^2(x), \dots, f^{\ell-1}(x)$ 两两不同

$$\text{② } \{f^k(x) \mid k \in \mathbb{Z}\} = \{x, f(x), f^2(x), \dots, f^{\ell-1}(x)\}.$$

③ 对 $\forall j \in \mathbb{Z}_{\geq 0}$ 且 $0 \leq j < \ell-1$, 有: $f(f^j(x)) = f^{j+1}(x)$. $f(f^{\ell-1}(x)) = x$.

$$\text{Proof: } \because r \in \mathbb{Z}_{\geq 1} \text{ 且 } f^r(x) = x \quad \therefore r \in \{n \in \mathbb{Z}_{\geq 1} \mid f^n(x) = x\}$$

$$\therefore \{n \in \mathbb{Z}_{\geq 1} \mid f^n(x) = x\} \neq \emptyset. \quad \therefore l = \min\{n \in \mathbb{Z}_{\geq 1} \mid f^n(x) = x\} \in \mathbb{Z}_{\geq 1}.$$

$$\text{已知 } f^0 = \text{id}_X. \quad \text{假设 } \exists h, k \in \{0, 1, \dots, l-1\}, h < k, \text{ s.t.}$$

$$f^h(x) = f^k(x). \quad \because h < k \quad \therefore k-h \in \mathbb{Z}_{\geq 1}$$

$$\therefore f: X \rightarrow X \text{ 是一个双射} \quad \therefore f^{-1}: X \rightarrow X \text{ 也是一个双射}$$

$$\therefore f^{k-h}(x) = f^k(f^{-h}(x)) = f^k((f^{-1})^h(x))$$

$$\because f^h(x) = f^k(x). \quad \therefore (f^{-1})^h(f^h(x)) = (f^{-1})^h(f^k(x))$$

$$\therefore x = f^{-h}(f^k(x)) = f^{(-h)+k}(x) = f^{k-h}(x).$$

$$\therefore k-h \in \mathbb{Z}_{\geq 1} \text{ 且 } f^{k-h}(x) = x. \quad \therefore k-h \in \{n \in \mathbb{Z}_{\geq 1} \mid f^n(x) = x\}$$

$$\therefore h \geq 0 \quad \therefore -h \leq 0. \quad \therefore k-h \leq k \leq l-1 < l$$

$$\therefore l \leq k-h < l \quad \therefore \text{矛盾} \quad \therefore x, f(x), f^2(x), \dots, f^{l-1}(x) \text{ 两两不同.}$$

①得证.

$$\therefore l = \min\{n \in \mathbb{Z}_{\geq 1} \mid f^n(x) = x\} \in \mathbb{Z}_{\geq 1} \quad \therefore l \in \mathbb{Z}_{\geq 1} \text{ 且 } f^l(x) = x.$$

$$\text{对 } \forall k \in \mathbb{Z}, \quad \because k \in \mathbb{Z} \text{ 且 } l \in \mathbb{Z}_{\geq 1} \quad \therefore \exists \text{ 唯一的 } q, r \in \mathbb{Z}, \text{ s.t.}$$

$$k = ql + r, \text{ 其中 } 0 \leq r < l \quad \therefore r \in \{0, 1, 2, \dots, l-1\}$$

$$\therefore f^k(x) = f^{ql+r}(x) = \cancel{f^{ql}(f^r(x))} f^r(f^l(x)) = f^r(x).$$

$$\therefore f^k(x) = f^r(x) \in \{x, f(x), f^2(x), \dots, f^{l-1}(x)\} \quad \therefore \text{左边} \subseteq \text{右边}.$$

右边 $\subseteq$ 左边显然. $\therefore$ ②得证.

$$\text{对 } \forall j \in \mathbb{Z}_{\geq 0} \text{ 且 } 0 \leq j < l-1, \text{ 有: } f(f^j(x)) = f^{j+1}(x).$$

$$f(f^{l-1}(x)) = f^l(x) = x. \quad \text{③得证.} \quad \square$$

定理(n 阶置换的循环分解) $n \in \mathbb{Z}_{\geq 1}$ 且 $\sigma \in \mathfrak{S}_n$, 则 σ 分解为两两不交的循环的乘积

$$\sigma = (a_{1,1} \cdots a_{1,l_1}) (a_{2,1} \cdots a_{2,l_2}) \cdots (a_{m,1} \cdots a_{m,l_m}) \quad (m \in \mathbb{Z}_{\geq 1}, \text{ 对 } \forall i, l_i \in \mathbb{Z}_{\geq 1})$$

满足 $\sum_{i=1}^m l_i = n$. 而且这些循环是唯一的, 至多差一个重排.

(“至多差一个重排”的含义是: 将置换 σ 分解为这 m 个两两不交的循环的乘积时, 这 m 个循环作为独立的数学实体是完全确定的, 唯一可以改变的仅仅是它们在连乘式中的先后书写顺序).

$$\text{Proof: } \because \sigma \in \mathfrak{S}_n \quad \therefore \langle \sigma \rangle \leq \mathfrak{S}_n \text{ 且 } \langle \sigma \rangle = \{\sigma^k \mid k \in \mathbb{Z}\}$$

$$\text{令 } G = \langle \sigma \rangle \quad \therefore G \leq \mathfrak{S}_n. \text{ 记 } X = \{1, 2, \dots, n\} \quad \because n \in \mathbb{Z}_{\geq 1}$$

$\therefore X$ 是非空有限集合. $\therefore \sigma: X \rightarrow X$ 是一个双射.

$$\text{定义映射 } \alpha: G \times X \longrightarrow X \\ (\tau, x) \longmapsto \tau(x)$$

$$\text{对 } \forall (\tau, x) \in G \times X, \text{ 有: } \tau \in \mathfrak{S}_n \text{ 且 } x \in \{1, 2, \dots, n\}.$$

$\therefore \tau$ 是 $\{1, \dots, n\} \rightarrow \{1, \dots, n\}$ 的双射 且 $x \in \{1, \dots, n\}. \therefore \tau(x) \in \{1, \dots, n\}.$

$\therefore a(\tau, x) = \tau(x) \in X$ ~~显然~~ 显然映射 a 把相等的自变量映到相等的映射值. $\therefore a$ 是 $G \times X \rightarrow X$ 的良定义的映射.

对 $\forall x \in X$, 有: $a(1_G, x) = a(1_{G_n}, x) = a(\text{id}_X, x)$
 $= \text{id}_X(x) = x$.

对 $\forall \tau_1, \tau_2 \in G$, ~~有: $\tau_1, \tau_2 \in G_n$~~ . $\forall x \in X = \{1, 2, \dots, n\}$, 有:

$\therefore \tau_1, \tau_2 \in G \quad \therefore \tau_1, \tau_2 \in G_n$.

$\therefore a(\tau_1, a(\tau_2, x)) = a(\tau_1, \tau_2(x)) = \tau_1(\tau_2(x)) = \tau_1\tau_2(x)$
 $= a(\tau_1\tau_2, x) \quad \therefore$ 映射 a 是群 G 在非空有限集合 X 上的群作用.

对 $\forall x \in X = \{1, 2, \dots, n\}$, 有:

$$Gx = \{a(\tau, x) \mid \tau \in G\} = \{\tau(x) \mid \tau \in G\} = \{\sigma^k(x) \mid k \in \mathbb{Z}\}$$

设在上述群作用下, $X = \{1, 2, \dots, n\}$ 的所有不同的轨道为

$C_1, C_2, \dots, C_m$ (其中 $m \in \mathbb{Z}_{\geq 1}$). $\therefore X = \{1, 2, \dots, n\} = C_1 \sqcup C_2 \sqcup \dots \sqcup C_m$

对每个 $i = 1, 2, \dots, m$, 从 C_i 中任取一个数 $a_{i,1}$ (即 $a_{i,1} \in C_i \subseteq X$).

$\therefore n \in \mathbb{Z}_{\geq 1}$ 且 $\sigma \in G_n \quad \therefore \sigma^{n!} = \text{id}_{\{1, \dots, n\}} = \text{id}_X \quad \therefore a_{i,1} \in X$

$\therefore n! \in \mathbb{Z}_{\geq 1}$ 且 $\sigma^{n!}(a_{i,1}) = \text{id}_X(a_{i,1}) = a_{i,1}$

$\therefore n! \in \{t \in \mathbb{Z}_{\geq 1} \mid \sigma^t(a_{i,1}) = a_{i,1}\} \quad \therefore \{t \in \mathbb{Z}_{\geq 1} \mid \sigma^t(a_{i,1}) = a_{i,1}\} \neq \emptyset$

$\therefore$ 由正整数集的非空子集必有最小元, ~~有~~

记 $l_i = \min\{t \in \mathbb{Z}_{\geq 1} \mid \sigma^t(a_{i,1}) = a_{i,1}\} \quad \therefore l_i \in \mathbb{Z}_{\geq 1}$ 且 $\sigma^{l_i}(a_{i,1}) = a_{i,1}$

定义 $a_{i,1} = a_{i,1} \in C_i \subseteq X$, $a_{i,2} = \sigma(a_{i,1}) \in X$,

$a_{i,3} = \sigma(a_{i,2}) = \sigma^2(a_{i,1}) \in X$, $a_{i,4} = \sigma(a_{i,3}) = \sigma^3(a_{i,1}) \in X$

.....

$a_{i,l_i-1} = \sigma^{l_i-2}(a_{i,1}) \in X$, $a_{i,l_i} = \sigma(a_{i,l_i-1}) = \sigma^{l_i-1}(a_{i,1}) \in X$

$\therefore \sigma(a_{i,l_i}) = \sigma^{l_i}(a_{i,1}) = a_{i,1}$

$\therefore X = \{1, 2, \dots, n\}$ ($n \in \mathbb{Z}_{\geq 1}$) 是非空有限集合, $\sigma: X \rightarrow X$ 是一个双射,

$a_{i,1} \in C_i \subseteq X$, $n! \in \mathbb{Z}_{\geq 1}$ 且 $\sigma^{n!}(a_{i,1}) = a_{i,1}$

$l_i = \min \{t \in \mathbb{Z}_{\geq 1} \mid \sigma^t(a_{i,1}) = a_{i,1}\}$

$\therefore a_{i,1}, \sigma(a_{i,1}) = a_{i,2}, \sigma^2(a_{i,1}) = a_{i,3}, \dots, \sigma^{l_i-1}(a_{i,1}) = a_{i,l_i}$ 两两不同,

且 $C_i = G_{a_{i,1}} = \{\sigma^k(a_{i,1}) \mid k \in \mathbb{Z}\} = \{a_{i,1}, a_{i,2}, a_{i,3}, \dots, a_{i,l_i}\}$

$\therefore n = |X| = \sum_{i=1}^m |C_i| = \sum_{i=1}^m l_i$

~~且~~ 对每个 $i=1, 2, \dots, m$, 记 $\gamma_i = (a_{i,1} \ a_{i,2} \ \dots \ a_{i,l_i}) \in \mathfrak{S}_X = \mathfrak{S}_n$

$\therefore \gamma_i(a_{i,1}) = a_{i,2} = \sigma(a_{i,1})$, $\gamma_i(a_{i,2}) = a_{i,3} = \sigma(a_{i,2})$

...

$\gamma_i(a_{i,l_i-1}) = a_{i,l_i} = \sigma(a_{i,l_i-1})$, $\gamma_i(a_{i,l_i}) = a_{i,1} = \sigma(a_{i,l_i})$

对 $\forall x \in X \setminus \{a_{i,1}, a_{i,2}, \dots, a_{i,l_i-1}, a_{i,l_i}\}$, 有: $\gamma_i(x) = x = \text{id}_X(x)$

$\therefore$ 对 $\forall x \in C_i$, 有: $\gamma_i(x) = \sigma(x)$

对 $\forall x \in X \setminus C_i$, 有: $\gamma_i(x) = x = \text{id}_X(x)$

$\therefore \gamma_1 \gamma_2 \dots \gamma_m \in \mathfrak{S}_X = \mathfrak{S}_n$.

对 $\forall x \in X = \{1, 2, \dots, n\}$. 存在唯一的 $i \in \{1, 2, \dots, m\}$, s.t. $x \in C_i$

$$\therefore (\gamma_1 \gamma_2 \cdots \gamma_m)(x) = \gamma_i(x) = \sigma(x)$$

$$\therefore \sigma = \gamma_1 \gamma_2 \cdots \gamma_m$$

$$= (a_{1,1} a_{1,2} \cdots a_{1,l_1}) (a_{2,1} a_{2,2} \cdots a_{2,l_2}) \cdots (a_{m,1} a_{m,2} \cdots a_{m,l_m})$$

$$\text{且有 } \sum_{i=1}^m l_i = n$$

$\therefore C_1, C_2, \dots, C_m$ 是 $X = \{1, 2, \dots, n\}$ 的两两不交的轨道

$\therefore \gamma_1, \gamma_2, \dots, \gamma_m \in \mathfrak{S}_X = \mathfrak{S}_n$ 是两两不交的循环. 存在性得证.

假设 $\sigma \in \mathfrak{S}_n$ 有两种方式分解为两两不交的循环的乘积:

$$\sigma = (a_{1,1} \cdots a_{1,l_1}) (a_{2,1} \cdots a_{2,l_2}) \cdots (a_{m,1} \cdots a_{m,l_m})$$

满足 $m \in \mathbb{Z}_{\geq 1}$ 且 $l_1 + l_2 + \cdots + l_m = n$ ($n \in \mathbb{Z}_{\geq 1}$) ($l_1, l_2, \dots, l_m \in \mathbb{Z}_{\geq 1}$)

$$\sigma = (b_{1,1} \cdots b_{1,t_1}) (b_{2,1} \cdots b_{2,t_2}) \cdots (b_{s,1} \cdots b_{s,t_s})$$

满足 $s \in \mathbb{Z}_{\geq 1}$ 且 $t_1 + t_2 + \cdots + t_s = n$ ($n \in \mathbb{Z}_{\geq 1}$) ($t_1, t_2, \dots, t_s \in \mathbb{Z}_{\geq 1}$)

对 $\forall i = 1, 2, \dots, m$, 定义

$$A_i = \{a_{i,1}, a_{i,2}, \dots, a_{i,l_i}\} \subseteq \{1, 2, \dots, n\} = X \quad \therefore A_i \neq \emptyset$$

$$\gamma_i = \{(a_{i,1}, a_{i,2}), (a_{i,2}, a_{i,3}), \dots, (a_{i,l_i-1}, a_{i,l_i}), (a_{i,l_i}, a_{i,1})\} \cup \{(x, x) \mid x \in X \setminus A_i\}$$

对 $\forall j = 1, 2, \dots, s$, 定义

$$B_j = \{b_{j,1}, b_{j,2}, \dots, b_{j,t_j}\} \subseteq \{1, 2, \dots, n\} = X \quad \therefore B_j \neq \emptyset.$$

$$d_j = \{(b_{j,1}, b_{j,2}), (b_{j,2}, b_{j,3}), \dots, (b_{j,t_j-1}, b_{j,t_j}), (b_{j,t_j}, b_{j,1})\} \cup \{(x, x) \mid x \in X \setminus B_j\}.$$

已知 $X = \{1, 2, \dots, n\} = A_1 \sqcup A_2 \sqcup \dots \sqcup A_m$

$$X = \{1, 2, \dots, n\} = B_1 \sqcup B_2 \sqcup \dots \sqcup B_s$$

对 $\forall x \in X$, 记 $O_\sigma(x) = \{\sigma^k(x) \mid k \in \mathbb{Z}\}$.

首先证明命题:

对 $\forall i = 1, 2, \dots, m$, $\forall x \in A_i$, 有: $A_i = O_\sigma(x)$

证明: $A_i = \{a_{i,1}, a_{i,2}, \dots, a_{i,\ell_i}\} \subseteq \{1, 2, \dots, n\} = X$.

对 $\forall y \in A_i$, $\because y \in A_i$ 且 $x \in A_i \quad \therefore \exists k \in \mathbb{Z}_{\geq 0}$, s.t. $y = \gamma_i^k(x)$

$\because x \in A_i \quad \therefore \gamma_i(x) = \sigma(x) \quad \therefore \gamma_i^k(x) = \sigma^k(x)$

$\therefore y = \gamma_i^k(x) = \sigma^k(x) \in O_\sigma(x) \quad \therefore A_i \subseteq O_\sigma(x)$

对 $\forall y \in O_\sigma(x)$, 有: $\exists k \in \mathbb{Z}$, s.t. $y = \sigma^k(x)$

$\because x \in A_i \quad \therefore \sigma(x) = \gamma_i(x) \quad \therefore \sigma^k(x) = \gamma_i^k(x)$

$\therefore y = \sigma^k(x) = \gamma_i^k(x) \in A_i \quad \therefore O_\sigma(x) \subseteq A_i \quad \therefore A_i = O_\sigma(x)$.

命题证毕.

再证明命题:

对 $\forall j = 1, 2, \dots, s$, $\forall x \in B_j$, 有: $B_j = O_\sigma(x)$

证明: $B_j = \{b_{j,1}, b_{j,2}, \dots, b_{j,t_j}\} \subseteq \{1, 2, \dots, n\} = X$

对 $\forall y \in B_j$, $\because y \in B_j \text{ 且 } x \in B_j \quad \therefore \exists k \in \mathbb{Z}_{\geq 0}$, s.t. $y = d_j^k(x)$

$\because x \in B_j \quad \therefore d_j^1(x) = \sigma(x) \quad \therefore d_j^k(x) = \sigma^k(x)$

$\therefore y = d_j^k(x) = \sigma^k(x) \in O_\sigma(x) \quad \therefore B_j \subseteq O_\sigma(x)$

对 $\forall y \in O_\sigma(x)$, 有: $\exists k \in \mathbb{Z}$, s.t. $y = \sigma^k(x)$

$\because x \in B_j \quad \therefore \sigma(x) = d_j^1(x) \quad \therefore \sigma^k(x) = d_j^k(x)$

$\therefore y = \sigma^k(x) = d_j^k(x) \in B_j \quad \therefore O_\sigma(x) \subseteq B_j \quad \therefore B_j = O_\sigma(x)$

命题证毕.

再证明命题: $\{A_1, A_2, \dots, A_m\} = \{B_1, B_2, \dots, B_s\}$ 且 $m = s$

证明: 对 $\forall i = 1, 2, \dots, m$, $\because A_i \neq \emptyset \quad \therefore$ 任取 $x \in A_i$

$\therefore A_i = O_\sigma(x) \quad \because x \in A_i \subseteq X = B_1 \sqcup B_2 \sqcup \dots \sqcup B_s$

$\therefore \exists j \in \{1, 2, \dots, s\}$, s.t. $x \in B_j \quad \therefore B_j = O_\sigma(x) \quad \therefore A_i = O_\sigma(x) = B_j$

$\therefore A_i \in \text{右边} \quad \therefore \text{左边} \subseteq \text{右边}$

对 $\forall j = 1, 2, \dots, s$, $\because B_j \neq \emptyset \quad \therefore$ 任取 $x \in B_j \quad \therefore B_j = O_\sigma(x)$

$\because x \in B_j \subseteq X = A_1 \sqcup A_2 \sqcup \dots \sqcup A_m \quad \therefore \exists i \in \{1, 2, \dots, m\}$, s.t. $x \in A_i$

$\therefore A_i = O_\sigma(x) \quad \therefore B_j = O_\sigma(x) = A_i \quad \therefore B_j \in \text{左边} \quad \therefore \text{右边} \subseteq \text{左边}$

$\therefore \text{左边} = \text{右边} \quad \therefore \{A_1, A_2, \dots, A_m\} = \{B_1, B_2, \dots, B_s\}$

$$\therefore m = |\{A_1, A_2, \dots, A_m\}| = |\{B_1, B_2, \dots, B_s\}| = s. \quad \text{命题证毕}$$

再证明命题: 对 $\forall i \in \{1, 2, \dots, m\}, \forall j \in \{1, 2, \dots, s\}$.

如果 $A_i = B_j$, 则有 $\gamma_i = \delta_j$.

证明: γ_i 和 δ_j 都是 $X \rightarrow X$ 的映射. $\because A_i = B_j \subseteq X$

$\therefore X \setminus A_i = X \setminus B_j$. 对 $\forall x \in X \setminus A_i$, 有: $x \in X \setminus A_i$ 且 $x \in X \setminus B_j$

$$\therefore \gamma_i(x) = x = \delta_j(x)$$

对 $\forall x \in A_i$, 有: $x \in A_i$ 且 $x \in B_j$. $\therefore \gamma_i(x) = \sigma(x) = \delta_j(x)$

$\therefore \gamma_i = \delta_j$ 命题得证.

唯一性得证.



命题 $Q(n)$: 任意一个 n -循环都可以分解成 $(n-1)$ 个对换的乘积. $n \in \mathbb{Z}_{\geq 1}$

Proof: 先看命题 $Q(1)$. (X 是非空集合).

设 σ 是一个任意的 1-循环. 设 $\sigma = (a_1)$, 其中 $a_1 \in X$.

$\therefore \sigma = \text{id}_X$ $\therefore \sigma$ 可以分解成 0 个对换的乘积. 命题 $Q(1)$ 得证.

再看命题 $Q(2)$.

设 σ 是一个任意的 2-循环. 设 $\sigma = (a_1 a_2)$, 其中 $a_1, a_2 \in X$ 且

$a_1 \neq a_2$. $\therefore \sigma$ 可以写成 1 个对换的乘积. 命题 $Q(2)$ 得证.

假设命题 $Q(k)$ 得证 ($k \in \mathbb{Z}_{\geq 2}$)

现在看命题 $Q(k+1)$.

设 σ 是一个任意的 $(k+1)$ -循环. 设 $\sigma = (a_1 a_2 \cdots a_k a_{k+1})$

(其中 $a_1, a_2, \dots, a_k, a_{k+1} \in X$ 且 $a_1, a_2, \dots, a_k, a_{k+1}$ 是 X 中相异的元素)

则有: $\sigma = (a_1 a_2 \cdots a_k a_{k+1}) = (a_1 a_{k+1})(a_1 a_2 \cdots a_k)$

$\therefore (a_1 a_2 \cdots a_k)$ 是 k -循环, $\therefore$ 由命题 $Q(k)$, $(a_1 a_2 \cdots a_k)$ 可以分解成 $(k-1)$ 个对换的乘积.

$\therefore \sigma = (a_1 a_{k+1})(a_1 a_2 \cdots a_k)$ 可以分解成 k 个对换的乘积.

$\therefore$ 命题 $Q(k+1)$ 得证.

$\therefore$ 对 $\forall n \in \mathbb{Z}_{\geq 1}$, 命题 $Q(n)$ 得证. $\square$

Remark: $(a_1 a_2 \cdots a_{n-1} a_n) = (a_1 a_n)(a_1 a_{n-1}) \cdots (a_1 a_3)(a_1 a_2)$
也可以很容易地用数学归纳法来证明. 此处略去.

~~定义~~

定义 (置换的支撑集). X 是非空集合, $\mathfrak{S}_X = \{\text{双射 } \sigma: X \rightarrow X\}$

已经证明了 $(\mathfrak{S}_X, \circ)$ 是群. 对 $\forall \sigma \in \mathfrak{S}_X$, 定义置换 σ 的支撑集 $\text{supp}(\sigma)$ 为

$\text{supp}(\sigma) = \{x \in X \mid \sigma(x) \neq x\} \subseteq X$.

Lemma: X 是非空集合 对 $\forall \sigma \in \mathfrak{S}_X$, 有: $\sigma(\text{supp}(\sigma)) = \text{supp}(\sigma)$.

Proof: $\because \sigma \in \mathfrak{S}_X \quad \therefore \text{supp}(\sigma) = \{x \in X \mid \sigma(x) \neq x\} \subseteq X$.

$$\therefore \sigma(\text{supp}(\sigma)) \subseteq X.$$

$$\text{对 } \forall y \in \sigma(\text{supp}(\sigma)), \exists x \in \text{supp}(\sigma), \text{ s.t. } y = \sigma(x)$$

$$\therefore x \in \text{supp}(\sigma) \quad \therefore x \in X \text{ 且 } \sigma(x) \neq x \quad \therefore x \in X \text{ 且 } \sigma: X \rightarrow X \text{ 是双射}$$

$$\therefore \sigma(x) \in X \quad \therefore y \in X. \quad \therefore y \in X, x \in X, y \neq x$$

$$\text{又 } \therefore \sigma: X \rightarrow X \text{ 是双射} \quad \therefore \sigma(y) \neq \sigma(x) \quad \therefore \sigma(y) \neq y$$

$$\therefore y \in X \text{ 且 } \sigma(y) \neq y \quad \therefore y \in \text{supp}(\sigma) \quad \therefore \sigma(\text{supp}(\sigma)) \subseteq \text{supp}(\sigma).$$

$$\text{对 } \forall y \in \text{supp}(\sigma), \text{ 有: } y \in \text{supp}(\sigma) \subseteq X \quad \therefore y \in X \text{ 且 } \sigma(y) \neq y$$

$$\therefore y \in X, \sigma: X \rightarrow X \text{ 是双射} \quad \therefore \exists x \in X, \text{ s.t. } \sigma(x) = y$$

$$\text{假设 } \sigma(x) = x. \text{ 则 } \therefore \sigma(x) = x, \sigma(x) = y \quad \therefore x = y \in X$$

$$\therefore \sigma: X \rightarrow X \text{ 是映射} \quad \therefore \sigma(x) = \sigma(y) \quad \therefore \sigma(y) = \sigma(x) = y$$

$$\therefore \sigma(y) = y \text{ 矛盾. } \therefore \sigma(x) \neq x.$$

$$\therefore x \in X \text{ 且 } \sigma(x) \neq x \quad \therefore x \in \text{supp}(\sigma) \subseteq X.$$

$$\therefore \sigma(x) \in \sigma(\text{supp}(\sigma)) \quad \therefore y \in \sigma(\text{supp}(\sigma))$$

$$\therefore \text{supp}(\sigma) \subseteq \sigma(\text{supp}(\sigma))$$

$$\therefore \sigma(\text{supp}(\sigma)) = \text{supp}(\sigma) \quad \square$$

Lemma: X 是非空集合, 对 $\forall \sigma, \tau \in \mathfrak{S}_X$, 若 $\text{supp}(\sigma) \cap \text{supp}(\tau) = \emptyset$, 则有: $\sigma \circ \tau = \tau \circ \sigma$

Proof: $\because \sigma \in \mathfrak{S}_X \quad \therefore \text{supp}(\sigma) = \{x \in X \mid \sigma(x) \neq x\} \subseteq X$

$\because \tau \in \mathfrak{S}_X \quad \therefore \text{supp}(\tau) = \{x \in X \mid \tau(x) \neq x\} \subseteq X$

$\because \sigma, \tau \in \mathfrak{S}_X, (\mathfrak{S}_X, \circ)$ 是群 $\therefore \sigma \circ \tau \in \mathfrak{S}_X, \tau \circ \sigma \in \mathfrak{S}_X$

对 $\forall x \in X$, 分三种情况讨论:

(i) $x \in \text{supp}(\sigma)$. 此时 $x \notin \text{supp}(\tau) \quad \therefore \tau(x) = x$

$\because x \in \text{supp}(\sigma) \quad \therefore \sigma(x) \in \sigma(\text{supp}(\sigma)) = \text{supp}(\sigma)$

$\therefore \sigma(x) \notin \text{supp}(\tau) \quad \because \sigma(x) \in X \quad \therefore \tau(\sigma(x)) = \sigma(x)$

$\therefore (\sigma \circ \tau)(x) = \sigma(\tau(x)) = \sigma(x) = \tau(\sigma(x)) = (\tau \circ \sigma)(x)$

(ii). $x \in \text{supp}(\tau)$. 此时 $x \notin \text{supp}(\sigma) \quad \therefore \sigma(x) = x$

$\because x \in \text{supp}(\tau) \subseteq X \quad \therefore \tau(x) \in \tau(\text{supp}(\tau)) = \text{supp}(\tau)$

$\therefore \tau(x) \notin \text{supp}(\sigma) \quad \because \tau(x) \in X \quad \therefore \sigma(\tau(x)) = \tau(x)$

$\therefore (\sigma \circ \tau)(x) = \sigma(\tau(x)) = \tau(x) = \tau(\sigma(x)) = (\tau \circ \sigma)(x)$

(iii) $x \in X$ 且 $x \notin \text{supp}(\sigma)$ 且 $x \notin \text{supp}(\tau)$

$\therefore \sigma(x) = x$ 且 $\tau(x) = x$

$\therefore (\sigma \circ \tau)(x) = \sigma(\tau(x)) = \sigma(x) = x = \tau(x) = \tau(\sigma(x)) = (\tau \circ \sigma)(x)$

$\therefore \sigma \circ \tau = \tau \circ \sigma \quad \square$

Lemma: X 是非空集合, $\sigma_1, \sigma_2, \dots, \sigma_m \in \mathcal{S}_X$ (其中 $m \in \mathbb{Z}_{\geq 1}$).

又对 $\forall i, j \in \{1, 2, \dots, m\}$ 且 $i \neq j$, 都有 $\text{supp}(\sigma_i) \cap \text{supp}(\sigma_j) = \emptyset$,

则有: $\sigma_1 \circ \sigma_2 \circ \dots \circ \sigma_m = \text{id}_X \iff \sigma_1 = \text{id}_X \text{ 且 } \sigma_2 = \text{id}_X \text{ 且 } \dots \text{ 且 } \sigma_m = \text{id}_X$.

Proof: ($\Leftarrow$): $\because \sigma_1 = \text{id}_X \text{ 且 } \sigma_2 = \text{id}_X \text{ 且 } \dots \text{ 且 } \sigma_m = \text{id}_X$

$$\therefore \sigma_1 \circ \sigma_2 \circ \dots \circ \sigma_m = \underbrace{\text{id}_X \circ \text{id}_X \circ \dots \circ \text{id}_X}_{m \uparrow \text{id}_X} = \text{id}_X.$$

($\Rightarrow$): 假设 $\exists k \in \{1, 2, \dots, m\}$, s.t. $\sigma_k \neq \text{id}_X$, 则有:

$\because \sigma_k \neq \text{id}_X \quad \therefore \exists x_0 \in X$, s.t. $\sigma_k(x_0) \neq x_0$. 记 $y_0 = \sigma_k(x_0)$

$\therefore y_0 \in X$ 且 $x_0 \in X$ 且 $y_0 \neq x_0$.

$\because x_0 \in X$ 且 $\sigma_k(x_0) \neq x_0 \quad \therefore x_0 \in \text{supp}(\sigma_k)$

$\because \sigma_k \in \mathcal{S}_X \quad \therefore \text{supp}(\sigma_k) \subseteq X \quad \therefore x_0 \in \text{supp}(\sigma_k) \subseteq X$

$\therefore \sigma_k(x_0) \in \sigma_k(\text{supp}(\sigma_k)) = \text{supp}(\sigma_k) \quad \therefore y_0 \in \text{supp}(\sigma_k)$

$\therefore x_0 \in \text{supp}(\sigma_k)$ 且 $y_0 \in \text{supp}(\sigma_k)$

$\therefore$ 对 $\forall i \in \{1, 2, \dots, m\} \setminus \{k\}$, 有: $\text{supp}(\sigma_i) \cap \text{supp}(\sigma_k) = \emptyset$

$\therefore$ 对 $\forall i \in \{1, 2, \dots, m\} \setminus \{k\}$, 有: $x_0 \notin \text{supp}(\sigma_i)$ 且 $y_0 \notin \text{supp}(\sigma_i)$

$\therefore$ 对 $\forall i \in \{1, 2, \dots, m\} \setminus \{k\}$, 有: $\sigma_i(x_0) = x_0$ 且 $\sigma_i(y_0) = y_0$

$\therefore x_0 = \text{id}_X(x_0) = (\sigma_1 \circ \sigma_2 \circ \dots \circ \sigma_m)(x_0)$

$$= (\sigma_1 \circ \dots \circ \sigma_{k-1}) \circ \sigma_k \circ (\sigma_{k+1} \circ \dots \circ \sigma_m)(x_0) = (\sigma_1 \circ \dots \circ \sigma_{k-1})(\sigma_k(x_0))$$

$$= (\sigma_1 \circ \dots \circ \sigma_{k-1})(y_0) = y_0 \quad \therefore y_0 = x_0 \quad \text{矛盾.}$$

$$\therefore \text{对 } \forall i \in \{1, 2, \dots, m\}, \text{ 都有 } \sigma_i = \text{id}_X$$

$$\therefore \sigma_1 = \text{id}_X \text{ 且 } \sigma_2 = \text{id}_X \text{ 且 } \dots \text{ 且 } \sigma_m = \text{id}_X. \quad \square$$

Lemma: X 是非空集合, 对 $\forall \sigma \in S_X$, $\forall k \in \mathbb{Z}$, 有:

$$\text{supp}(\sigma^k) \subseteq \text{supp}(\sigma).$$

Proof: 当 $k=0$ 时, $\sigma^k = \sigma^0 = \text{id}_X$. $\therefore \text{supp}(\sigma^k) = \text{supp}(\text{id}_X) = \emptyset$

显然有 $\text{supp}(\sigma^k) \subseteq \text{supp}(\sigma)$

当 $k \in \mathbb{Z}_{\geq 1}$ 时, 对 $\forall x \in \text{supp}(\sigma^k)$, 有: $x \in X$ 且 $\sigma^k(x) \neq x$.

假设 $x \notin \text{supp}(\sigma)$, 则 $\sigma(x) = x$.

$$\therefore \sigma^k(x) = (\underbrace{\sigma \circ \dots \circ \sigma}_{k \text{ 个 } \sigma})(x) = x \quad \text{矛盾.} \quad \therefore x \in \text{supp}(\sigma)$$

$$\therefore \text{supp}(\sigma^k) \subseteq \text{supp}(\sigma)$$

当 $k \in \mathbb{Z}_{\leq -1}$ 时, $-k \in \mathbb{Z}_{\geq 1}$ 且 $k = -(-k)$

$$\therefore \sigma^k = \sigma^{-(-k)} = (\sigma^{-1})^{-k}$$

$$\therefore \text{supp}(\sigma^k) = \text{supp}((\sigma^{-1})^{-k}) \subseteq \text{supp}(\sigma^{-1}) = \text{supp}(\sigma)$$

$$\therefore \text{对 } \forall k \in \mathbb{Z}, \text{ 有: } \text{supp}(\sigma^k) \subseteq \text{supp}(\sigma) \quad \square$$

Lemma: X 是非空集合, $\sigma, \tau \in \mathfrak{S}_X$, $\text{supp}(\sigma) \cap \text{supp}(\tau) = \emptyset$,

则有: 对 $\forall k, l \in \mathbb{Z}$, 有: $\sigma^k \circ \tau^l = \tau^l \circ \sigma^k$

proof: 对 $\forall k, l \in \mathbb{Z}$, 有:

$$\because \sigma \in \mathfrak{S}_X \quad \therefore \sigma^k \in \mathfrak{S}_X \quad \because \tau \in \mathfrak{S}_X \quad \therefore \tau^l \in \mathfrak{S}_X.$$

$$\because \sigma \in \mathfrak{S}_X, k \in \mathbb{Z} \quad \therefore \text{supp}(\sigma^k) \subseteq \text{supp}(\sigma)$$

$$\because \tau \in \mathfrak{S}_X, l \in \mathbb{Z} \quad \therefore \text{supp}(\tau^l) \subseteq \text{supp}(\tau)$$

$$\because \text{supp}(\sigma) \cap \text{supp}(\tau) = \emptyset \quad \therefore \text{supp}(\sigma^k) \cap \text{supp}(\tau^l) = \emptyset$$

$$\therefore \sigma^k \circ \tau^l = \tau^l \circ \sigma^k$$



Lemma: $n \in \mathbb{Z}_{\geq 1}$ 且 $\sigma \in \mathfrak{S}_n$, 则已经证明] 存在唯一的 σ 分解为两两不交的循环的乘积的方式:

$$\sigma = (a_{1,1} \cdots a_{1,l_1}) (a_{2,1} \cdots a_{2,l_2}) \cdots (a_{m,1} \cdots a_{m,l_m})$$

$$\text{满足 } \sum_{i=1}^m l_i = n.$$

$$\text{则有: 对 } \forall k \in \mathbb{Z}, \text{ 有: } \sigma^k = (a_{1,1} \cdots a_{1,l_1})^k (a_{2,1} \cdots a_{2,l_2})^k \cdots (a_{m,1} \cdots a_{m,l_m})^k$$

$$\text{Proof: 当 } k=0 \text{ 时, 左边} = \sigma^0 = \text{id}_X. \text{ 右边} = \underbrace{\text{id}_X \circ \text{id}_X \circ \cdots \circ \text{id}_X}_{m \uparrow \text{id}_X} = \text{id}_X$$

$$\therefore \text{左边} = \text{右边}.$$

$$\text{当 } k \in \mathbb{Z}_{\geq 1} \text{ 时, } \sigma^k = \left((a_{1,1} \cdots a_{1,l_1}) (a_{2,1} \cdots a_{2,l_2}) \cdots (a_{m,1} \cdots a_{m,l_m}) \right)^k$$

$$\text{supp}((a_{1,1} \cdots a_{1,l_1})) = \{a_{1,1}, \dots, a_{1,l_1}\} \subseteq \{1, 2, \dots, n\}$$

$$\text{supp}((a_{2,1} \cdots a_{2,l_2})) = \{a_{2,1}, \dots, a_{2,l_2}\} \subseteq \{1, 2, \dots, n\}$$

.....

$$\text{supp}((a_{m,1} \cdots a_{m,l_m})) = \{a_{m,1}, \dots, a_{m,l_m}\} \subseteq \{1, 2, \dots, n\}.$$

$$\therefore \sigma^k = (a_{1,1} \cdots a_{1,l_1})^k (a_{2,1} \cdots a_{2,l_2})^k \cdots (a_{m,1} \cdots a_{m,l_m})^k$$

(只需有限次交换即可).

当 $k \in \mathbb{Z}_{\leq -1}$ 时, $-k \in \mathbb{Z}_{\geq 1}$. $k = -(-k)$.

$$\therefore \sigma^k = \sigma^{-(-k)} = (\sigma^{-1})^{-k} = \left((a_{m,1} \cdots a_{m,l_m})^{-1} \cdots (a_{2,1} \cdots a_{2,l_2})^{-1} (a_{1,1} \cdots a_{1,l_1})^{-1} \right)^{-k}$$

$$= \left((a_{m,1} \cdots a_{m,l_m})^{-1} \right)^{-k} \cdots \left((a_{2,1} \cdots a_{2,l_2})^{-1} \right)^{-k} \left((a_{1,1} \cdots a_{1,l_1})^{-1} \right)^{-k}$$

$$= (a_{m,1} \cdots a_{m,l_m})^k \cdots (a_{2,1} \cdots a_{2,l_2})^k (a_{1,1} \cdots a_{1,l_1})^k$$

$$= (a_{1,1} \cdots a_{1,l_1})^k (a_{2,1} \cdots a_{2,l_2})^k \cdots (a_{m,1} \cdots a_{m,l_m})^k \quad (\text{有限次交换即可})$$

□

~~定理~~ 定理 (用置换的循环分解来表示置换的阶和符号). $n \in \mathbb{Z}_{\geq 1}$ 且 $\sigma \in \mathbb{S}_n$,

则已经证明了存在唯一的 σ 分解为两两不交的循环的乘积的方式:

$$\sigma = (a_{1,1} \cdots a_{1,l_1}) (a_{2,1} \cdots a_{2,l_2}) \cdots (a_{m,1} \cdots a_{m,l_m})$$

$$\text{满足 } \sum_{i=1}^m l_i = n, \quad m \in \mathbb{Z}_{\geq 1}, \quad l_1 \in \mathbb{Z}_{\geq 1}, \quad l_2 \in \mathbb{Z}_{\geq 1}, \quad \dots, \quad l_m \in \mathbb{Z}_{\geq 1}$$

$$\text{则有: } \text{sgn}(\sigma) = (-1)^{\sum_{i=1}^m (l_i - 1)}$$

$$\text{ord}(\sigma) = \text{lcm}(l_1, \dots, l_m)$$

Proof: ~~对~~ 对 $\forall i=1, 2, \dots, m$, 记 $\xi_i = (a_{i,1} \dots a_{i,l_i})$

$$\therefore \sigma = \xi_1 \xi_2 \dots \xi_m \quad \because \sigma \in S_n, \quad \xi_1 \in S_n, \quad \xi_2 \in S_n, \quad \dots, \quad \xi_m \in S_n$$

$$\therefore \operatorname{sgn}(\sigma) = \operatorname{sgn}(\xi_1 \xi_2 \dots \xi_m) = \operatorname{sgn}(\xi_1) \operatorname{sgn}(\xi_2) \dots \operatorname{sgn}(\xi_m)$$

$$\text{对 } \forall i=1, 2, \dots, m, \quad \because \xi_i = (a_{i,1} \dots a_{i,l_i})$$

$\therefore \xi_i$ 可以分解成 $(l_i - 1)$ 个对换的乘积.

$$\therefore \xi_i = T_1 T_2 \dots T_{l_i-1}$$

$$\therefore \operatorname{sgn}(\xi_i) = \operatorname{sgn}(T_1 T_2 \dots T_{l_i-1}) = \operatorname{sgn}(T_1) \operatorname{sgn}(T_2) \dots \operatorname{sgn}(T_{l_i-1})$$

$$= \underbrace{(-1) \times (-1) \times \dots \times (-1)}_{(l_i-1)\uparrow} = (-1)^{l_i-1}$$

$$(\text{如果 } l_i = 1, \text{ 则 } \xi_i = \operatorname{id}_X = \operatorname{id}_{\{1,2,\dots,n\}} \quad \therefore l(\xi_i) = 0)$$

$$\therefore \operatorname{sgn}(\xi_i) = \operatorname{sgn}(\operatorname{id}_X) = (-1)^0 = 1, \text{ 符合上面的结果.}$$

$$\therefore \operatorname{sgn}(\sigma) = \prod_{i=1}^m \operatorname{sgn}(\xi_i) = \prod_{i=1}^m (-1)^{l_i-1} = (-1)^{\sum_{i=1}^m (l_i-1)}$$

对 $\forall k \in \mathbb{Z}$, 有:

$$\sigma^k = \operatorname{id}_{\{1,2,\dots,n\}} \Leftrightarrow \xi_1^k \xi_2^k \dots \xi_m^k = \operatorname{id}_{\{1,2,\dots,n\}}$$

$$\Leftrightarrow \xi_1^k = \operatorname{id}_X \text{ 且 } \xi_2^k = \operatorname{id}_X \text{ 且 } \dots \text{ 且 } \xi_m^k = \operatorname{id}_X$$

$$\Leftrightarrow \operatorname{ord}(\xi_1) \mid k \text{ 且 } \operatorname{ord}(\xi_2) \mid k \text{ 且 } \dots \text{ 且 } \operatorname{ord}(\xi_m) \mid k$$

$$\Leftrightarrow l_1 \mid k \text{ 且 } l_2 \mid k \text{ 且 } \dots \text{ 且 } l_m \mid k$$

$$\therefore \operatorname{ord}(\sigma) = \min \{ k \in \mathbb{Z}_{\geq 1} \mid \sigma^k = \operatorname{id}_{\{1,2,\dots,n\}} \} = \min \{ k \in \mathbb{Z}_{\geq 1} \mid l_1 \mid k \text{ 且 } l_2 \mid k \text{ 且 } \dots \text{ 且 } l_m \mid k \}$$

$$= \text{lcm}(l_1, l_2, \dots, l_m) \quad \square$$

Remark: $\because n \in \mathbb{Z}_{\geq 1} \quad \therefore X = \{1, 2, \dots, n\}$ 是非空集合

$$\because k \in \mathbb{Z} \quad \therefore \xi_1^k \in G_n, \xi_2^k \in G_n, \dots, \xi_m^k \in G_n$$

对 $\forall i, j \in \{1, 2, \dots, m\}$ 且 $i \neq j$, ~~ξ_i^k~~

$$\therefore \text{supp}(\xi_i) \cap \text{supp}(\xi_j) = \emptyset, \quad \text{supp}(\xi_i^k) \subseteq \text{supp}(\xi_i)$$

$$\text{supp}(\xi_j^k) \subseteq \text{supp}(\xi_j) \quad \therefore \text{supp}(\xi_i^k) \cap \text{supp}(\xi_j^k) = \emptyset$$

$$\therefore \xi_1^k \cdot \xi_2^k \cdots \xi_m^k = \text{id}_X \iff \xi_1^k = \text{id}_X \text{ 且 } \xi_2^k = \text{id}_X \text{ 且 } \cdots \text{ 且 } \xi_m^k = \text{id}_X$$

Lemma: G 是群, $A \leq G$, A 是交换群, $G = Z_G A$, 则有:

G 是交换群.

$$\text{Proof: } \because G \text{ 是群} \quad \therefore Z_G = \{x \in G \mid \text{对 } \forall g \in G, \text{ 都有 } xg = gx\}$$

已经证明了 $Z_G \leq G$, $Z_G \triangleleft G$

$$\because G \text{ 是群}, A \leq G, Z_G \triangleleft G \quad \therefore AZ_G = Z_G A \text{ 且 } AZ_G \leq G$$

$$\therefore Z_G A \leq G$$

$$\text{对 } \forall x, y \in G, \quad \because x \in G = Z_G A \quad \therefore \exists z_1 \in Z_G, a_1 \in A, \text{ s.t.}$$

$$x = z_1 a_1. \quad \because y \in G = Z_G A \quad \therefore \exists z_2 \in Z_G, a_2 \in A, \text{ s.t. } y = z_2 a_2$$

$$\therefore xy = (z_1 a_1)(z_2 a_2) = z_1 (a_1 z_2) a_2 = z_1 (z_2 a_1) a_2 = (z_1 z_2)(a_1 a_2)$$

$$= (z_2 z_1)(a_2 a_1) = z_2 (z_1 a_2) a_1 = z_2 (a_2 z_1) a_1 = (z_2 a_2)(z_1 a_1)$$

$$= yx \quad \therefore G \text{ 是交换群.} \quad \square$$

Lemma: G 是群, $N \triangleleft G$, $a \in G$, 则对 $\forall n \in \mathbb{Z}$, 都有

$$(aN)^n = a^n N$$

Proof: $\because G$ 是群, $N \triangleleft G \therefore G/N$ 是群, $\because a \in G \therefore aN \in G/N$.

当 $n=0$ 时, $(aN)^0 = 1_{G/N} = 1_G N = a^0 N \therefore$ 当 $n=0$ 时结论成立.

对 $\forall n \in \mathbb{Z}_{\geq 1}$, 有: $(aN)^n = \underbrace{(aN) \cdot (aN) \cdots (aN)}_{n \uparrow aN} = \cancel{a^n N} a^n N$

~~对 $\forall n \in \mathbb{Z}_{\geq 1}$~~ $\therefore$ 对 $\forall n \in \mathbb{Z}_{\geq 1}$ 结论成立.

对 $\forall n \in \mathbb{Z}_{\leq -1}$, 有: $-n \in \mathbb{Z}_{\geq 1}$ 且 $n = -(-n)$

$$\begin{aligned} \therefore (aN)^n &= (aN)^{-(-n)} = ((aN)^{-1})^{-n} = (a^{-1}N)^{-n} = (a^{-1})^{-n} N = a^{-(-n)} N \\ &= a^n N \end{aligned} \therefore \text{对 } \forall n \in \mathbb{Z}_{\leq -1}, \text{ 结论成立. } \square$$

Lemma: G 是群, G/Z_G 是循环群, 则有: G 是交换群.

Proof: $\because G$ 是群 $\therefore Z_G = \{x \in G \mid \text{对 } \forall g \in G, \text{ 都有 } xg = gx\}$.

已经证明了 $Z_G \leq G$, $Z_G \triangleleft G \therefore G/Z_G$ 是群.

$\because G/Z_G$ 是循环群 $\therefore \exists aZ_G \in G/Z_G$ (其中 $a \in G$), s.t.

$$G/Z_G = \langle aZ_G \rangle \quad \because a \in G \quad \therefore \text{记 } A = \langle a \rangle \leq G \quad \therefore A \leq G$$

$\therefore A = \langle a \rangle \quad \therefore A$ 是循环群 $\therefore A$ 是交换群.

$\because G$ 是群, $A \leq G$, $Z_G \triangleleft G \quad \therefore AZ_G = Z_G A$ 且 $AZ_G \leq G$

$\therefore Z_G A \leq G$

对 $\forall g \in G$, $\because g \in G \quad \therefore gZ_G \in G/Z_G \quad \therefore gZ_G \in \langle aZ_G \rangle$

$\therefore \exists k \in \mathbb{Z}$, s.t. $gZ_G = (aZ_G)^k = a^k Z_G$

$\therefore g = g \cdot 1_G \in gZ_G = a^k Z_G \quad \therefore \exists \zeta \in Z_G$, s.t. $g = a^k \zeta$

$\because \zeta \in Z_G$, $a^k \in G \quad \therefore \zeta a^k = a^k \zeta \quad \therefore g = \zeta a^k$

$\because a \in G$, $k \in \mathbb{Z} \quad \therefore a^k \in \langle a \rangle = A \quad \therefore g = \zeta a^k \in Z_G A$

$\therefore G \subseteq Z_G A \leq G \quad \therefore G = Z_G A$

$\because G$ 是群, $A \leq G$, A 是交换群, $G = Z_G A \quad \therefore G$ 是交换群. $\square$

Recall Lemma: ~~素数阶群~~

G 是有限群, $|G|$ 是素数, 则 G 是循环群, G 也是交换群.

Lemma: G 是有限群, $|G| = p^2$ (p 是素数), 则有: G 是交换群.

Proof: $\because G$ 是有限群, p 是素数, $|G| = p^2 \quad \therefore G$ 是 p -群.

$\because p \in \mathbb{Z}_{\geq 2} \quad \therefore p^2 \geq 2^2 = 4 \quad \therefore |G| \geq 4 \quad \therefore G$ 是非平凡的 p -群

$\therefore \{1_G\} \subsetneq Z_G \triangleleft G \quad \because G$ 是有限群, $Z_G \triangleleft G \quad \therefore |Z_G| \mid |G|$

$\therefore |Z_G| \mid p^2 \quad |Z_G| > 1 \quad \therefore |Z_G| \geq 2 \quad \therefore |Z_G| = p$ 或 $|Z_G| = p^2$

假设 $|Z_G| = p$, 则有: $|G| = |Z_G| (G:Z_G) \therefore p^2 = p \cdot (G:Z_G)$

$\therefore (G:Z_G) = p \quad \therefore |G/Z_G| = (G:Z_G) = p \quad \therefore G/Z_G$ 是循环群

$\therefore G$ 是交换群 $\therefore Z_G = G \quad \therefore |Z_G| = |G| = p^2 \quad \therefore p = p^2$

$\therefore p = 0$ 或 $p = 1$. 矛盾. $\therefore |Z_G| \neq p$

$\therefore |Z_G| = p^2 \quad \therefore |Z_G| = |G|$ 且 $Z_G \triangleleft G$, G 是有限群

$\therefore G = Z_G \quad \therefore G$ 是交换群. $\square$